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Course: Artificial Intelligence

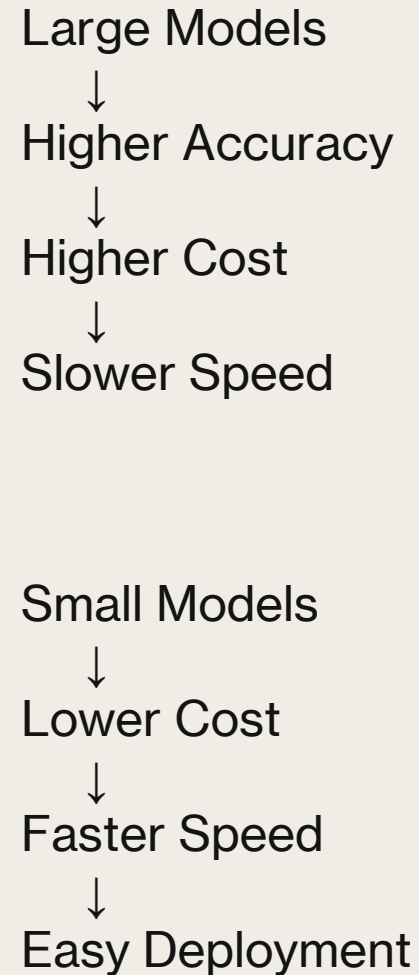
****Artifact 10: Pre-Trained Models Decision Matrix**



PRE-TRAINED MODEL TRADE-OFF ANALYSIS

WHY MODEL SELECTION MATTERS

- Organizations use different AI models depending on their requirements.
- When selecting a model, engineers must balance:
- Accuracy
- Speed
- Model Size
- Energy Consumption
- Hardware Requirements
- Explainability



AI Model Comparison			
Characteristic	Language Models	Image Generation Models	Other Models
Use case	Text generation	Image creation	Vision or speech
Model size	Larger is more powerful	Larger is more powerful	Larger is more powerful
Performance vs cost	Good for chat interface	Smaller models may be enough	Smaller models may be enough
Availability	Check maintenance and download ease	Check maintenance and download ease	Check maintenance and download ease

SELECTED PRE-TRAINED MODELS

NLP / Generative AI

BERT

Developed by Google

110 Million Parameters

Excellent for text classification

GPT-4

Advanced Generative AI

Trillions of parameters (estimated)

Excellent language generation

Computer Vision

MobileNetV3

Lightweight CNN

Designed for mobile devices

EfficientNet-B0

Balanced speed and accuracy

Tabular Data

LightGBM

Fast gradient boosting

XGBoost

Industry-leading tabular model



DECISION MATRIX

Model	Domain	Size	Accuracy	Speed	Explainability
BERT	NLP	Medium	High	Medium	Medium
GPT-4	Generative AI	Very Large	Very High	Slow	Low
MobileNetV3	Vision	Small	Good	Very Fast	Medium
EfficientNet-B0	Vision	Small-Medium	High	Fast	Medium
LightGBM	Tabular	Small	High	Very Fast	High
XGBoost	Tabular	Medium	Very High	Fast	High

Decision-Making Pugh Matrix

The Pugh Matrix is a helpful tool for comparing and prioritizing different options based on multiple criteria. The scores in the matrix can guide decision-making by highlighting the relative strengths and weaknesses of each option. Learn more at www.ignitecorporation.com/pugh-matrix



Criteria	Option A	Option B	Option C	Option D
Cost-effectiveness	+	0	-	0
Market Potential	-	+	0	0
Innovation	0	0	+	-
Scalability	0	-	+	-
Customer Satisfaction	+	-	0	0
Total	1	0	-1	0

Key Notes: (+) better option
(-) worse option
(0) equal option



ANALYSIS AND RECOMMENDATIONS

Strengths and Recommendations

GPT-4

Strengths

Best language generation

Strong reasoning

Weaknesses

Expensive

Slow inference

Low explainability

Recommended for:

Chatbots

Content generation

Virtual assistants

BERT

Strengths

Strong language understanding

Lower cost than GPT

Recommended for:

Classification

Sentiment analysis

MobileNetV3

Strengths

Very fast

Small size

Recommended for:

Smartphones

Edge AI

LightGBM and XGBoost

Strengths

Excellent for structured data

High interpretability

Recommended for:

Finance

Healthcare

Fraud detection

EfficientNet-B0

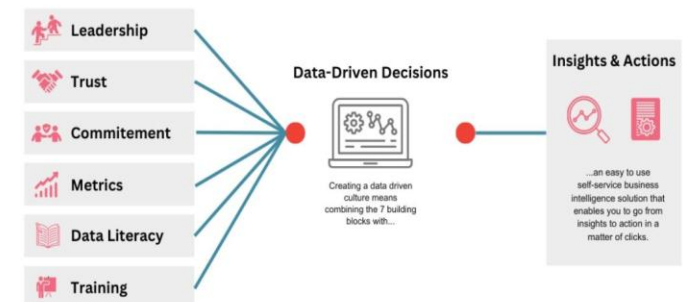
Strengths

Balanced

performance

Recommended for

Data-Driven Decision Making



CONCLUSION

KEY FINDINGS

- ✓ NO SINGLE MODEL IS BEST FOR EVERY TASK.
- ✓ LARGER MODELS PROVIDE BETTER ACCURACY BUT REQUIRE MORE RESOURCES.
- ✓ SMALLER MODELS OFFER FASTER DEPLOYMENT AND LOWER COST.
- ✓ EXPLAINABILITY REMAINS CRITICAL IN REGULATED INDUSTRIES.
- ✓ MODEL SELECTION SHOULD BALANCE PERFORMANCE, COST, SPEED, AND TRANSPARENCY.

Task	Recommended Model
Chatbot	GPT-4
Text Classification	BERT
Mobile Vision	MobileNetV3
Image Classification	EfficientNet-B0
Structured Data Analytics	LightGBM
Business Prediction Models	XGBoost

